

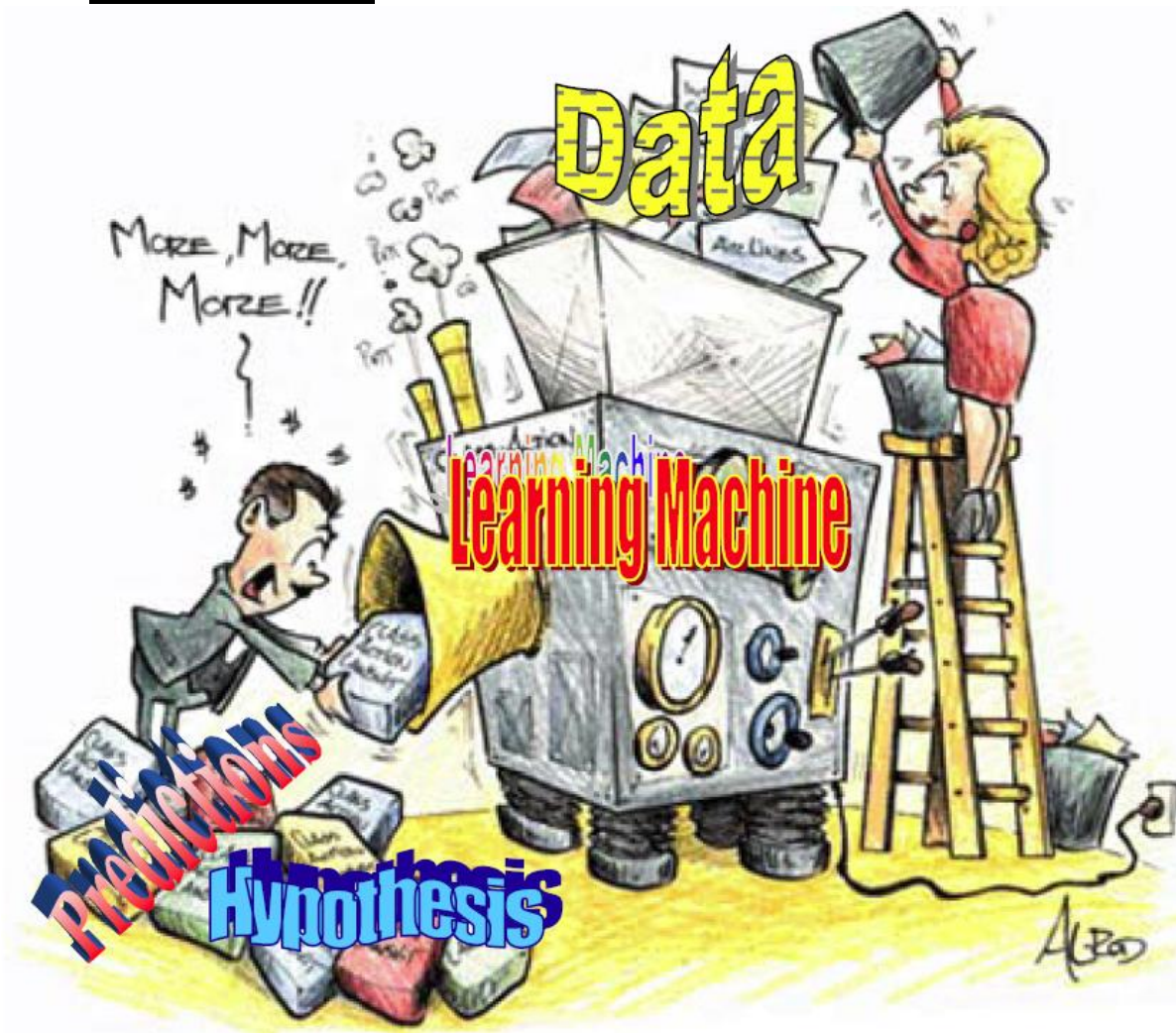
Introduction

Le Song

Computing for Data Analysis
(Introduction to Machine Learning)
CX 4240, Spring 2014

What is Machine Learning (ML)

- Study of algorithms that improve their performance at some task with experience



Common to Industrial scale problems



13 million wikipedia pages



facebook

800 million users



flickr™

6 billion photos



You Tube

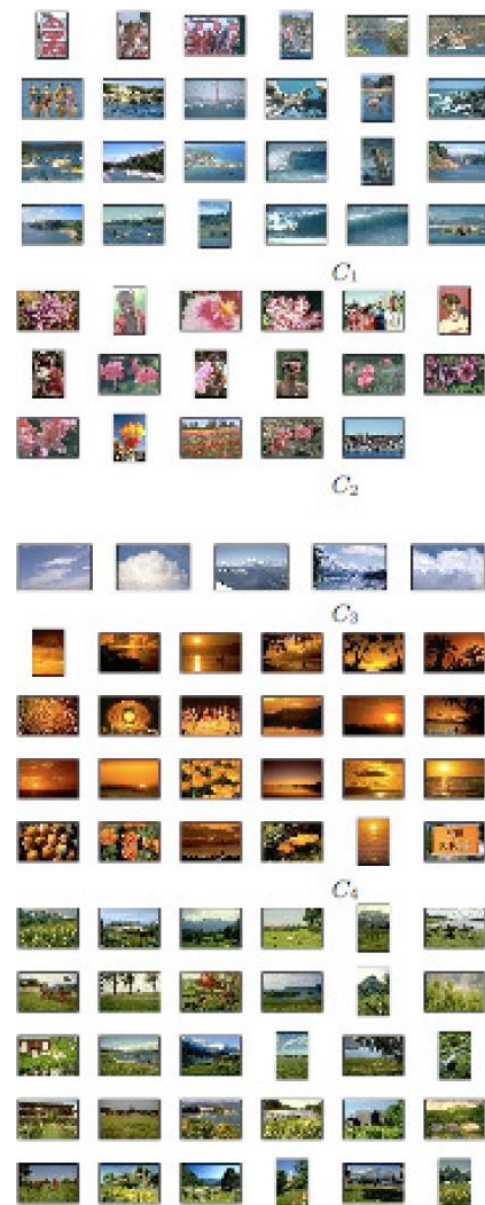
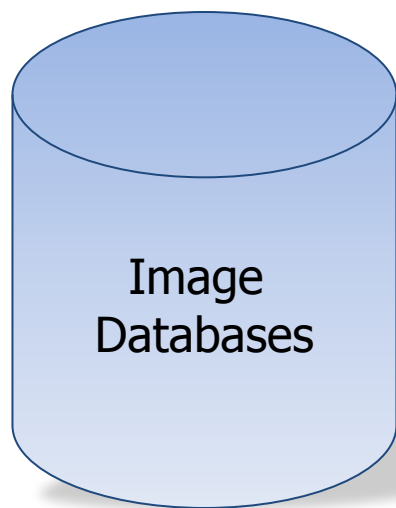
24 hours video uploaded per minutes



Google

> 1 trillion webpages

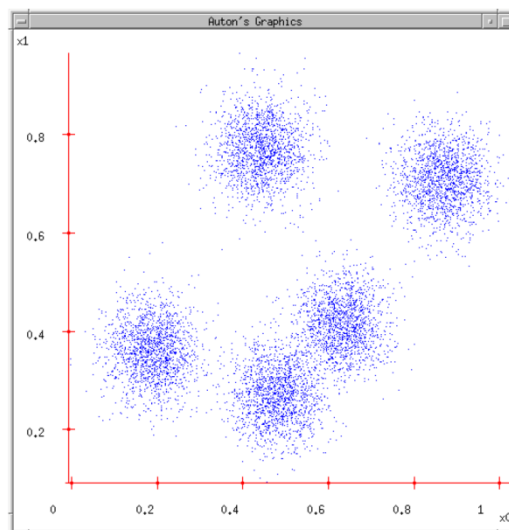
Organizing Images



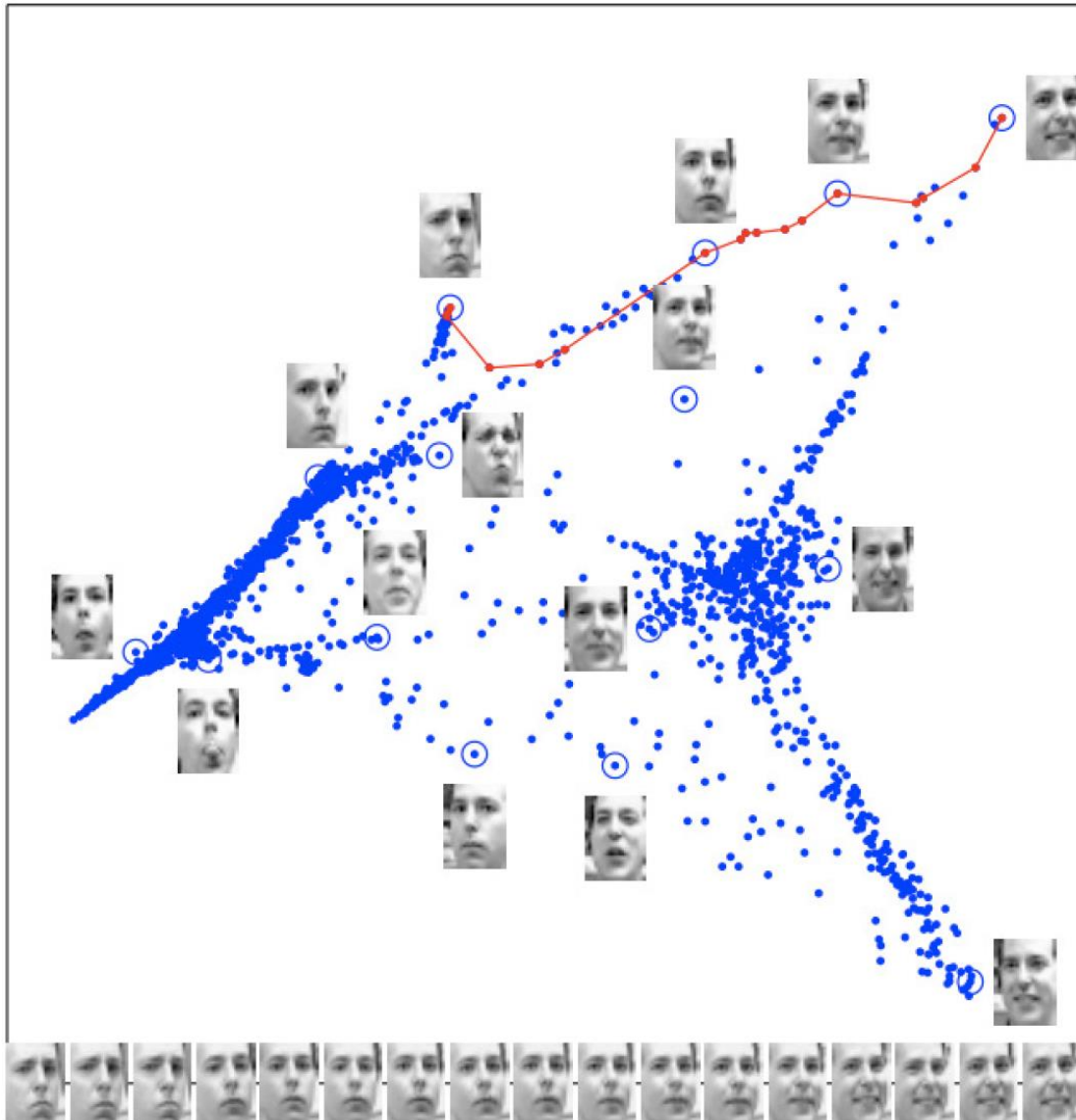
What are the desired outcomes?

What are the inputs (data)?

What are the learning paradigms?



Visualize Image Relations



Each image has thousands or millions of pixels.

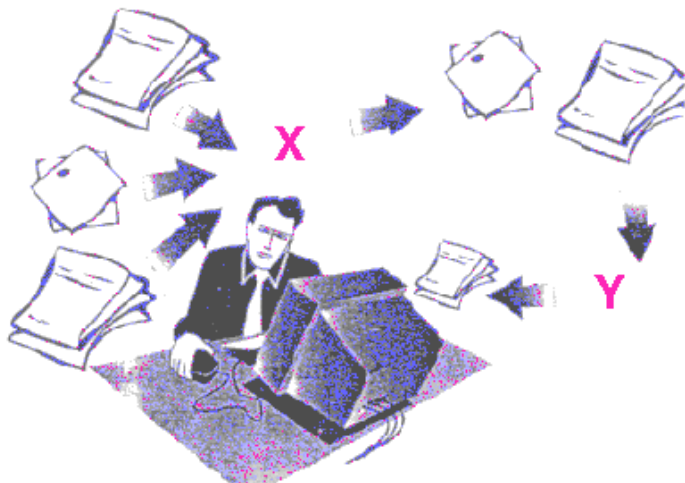
What are the desired outcomes?

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Organizing documents

- Reading, digesting, and categorizing a vast text database is too much for human!



- We want:

"Arts"	"Budgets"	"Children"	"Education"
NEW	MILLION	CHILDREN	SCHOOL
FILM	TAX	WOMEN	STUDENTS
SHOW	PROGRAM	PEOPLE	SCHOOLS
MUSIC	BUDGET	CHILD	EDUCATION
MOVIE	BILLION	YEARS	TEACHERS
PLAY	FEDERAL	FAMILIES	HIGH
MUSICAL	YEAR	WORK	PUBLIC
BEST	SPENDING	PARENTS	TEACHER
ACTOR	NEW	SAYS	BENNETT
FIRST	STATE	FAMILY	MANIGAT
YORK	PLAN	WELFARE	NAMPHY
OPERA	MONEY	MEN	STATE
THEATER	PROGRAMS	PERCENT	PRESIDENT
ACTRESS	GOVERNMENT	CARE	ELEMENTARY
LOVE	CONGRESS	LIFE	HAITI

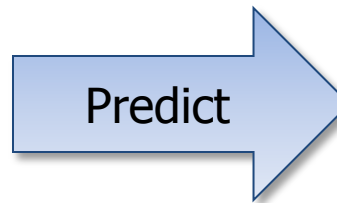
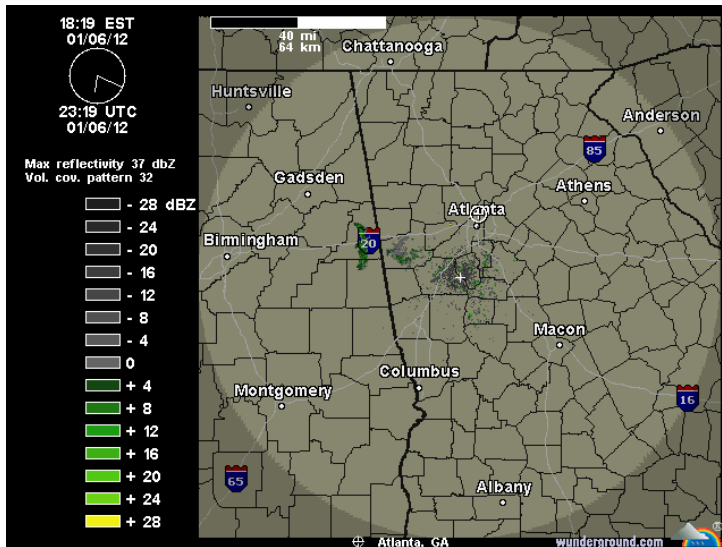
The William Randolph Hearst Foundation will give \$1.25 million to Lincoln Center, Metropolitan Opera Co., New York Philharmonic and Juilliard School. "Our board felt that we had a real opportunity to make a mark on the future of the performing arts with these grants an act every bit as important as our traditional areas of support in health, medical research, education and the social services," Hearst Foundation President Randolph A. Hearst said Monday in announcing the grants. Lincoln Center's share will be \$200,000 for its new building, which will house young artists and provide new public facilities. The Metropolitan Opera Co. and New York Philharmonic will receive \$400,000 each. The Juilliard School, where music and the performing arts are taught, will get \$250,000. The Hearst Foundation, a leading supporter of the Lincoln Center Consolidated Corporate Fund, will make its usual annual \$100,000 donation, too.

What are the desired outcomes?

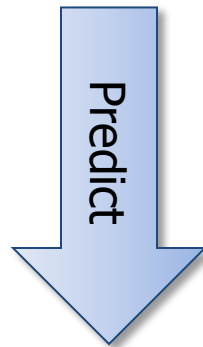
What are the inputs (data)?

What are the learning paradigms?

Weather Prediction



Numeric values:
40 F
Wind: NE at 14 km/h
Humidity: 83%

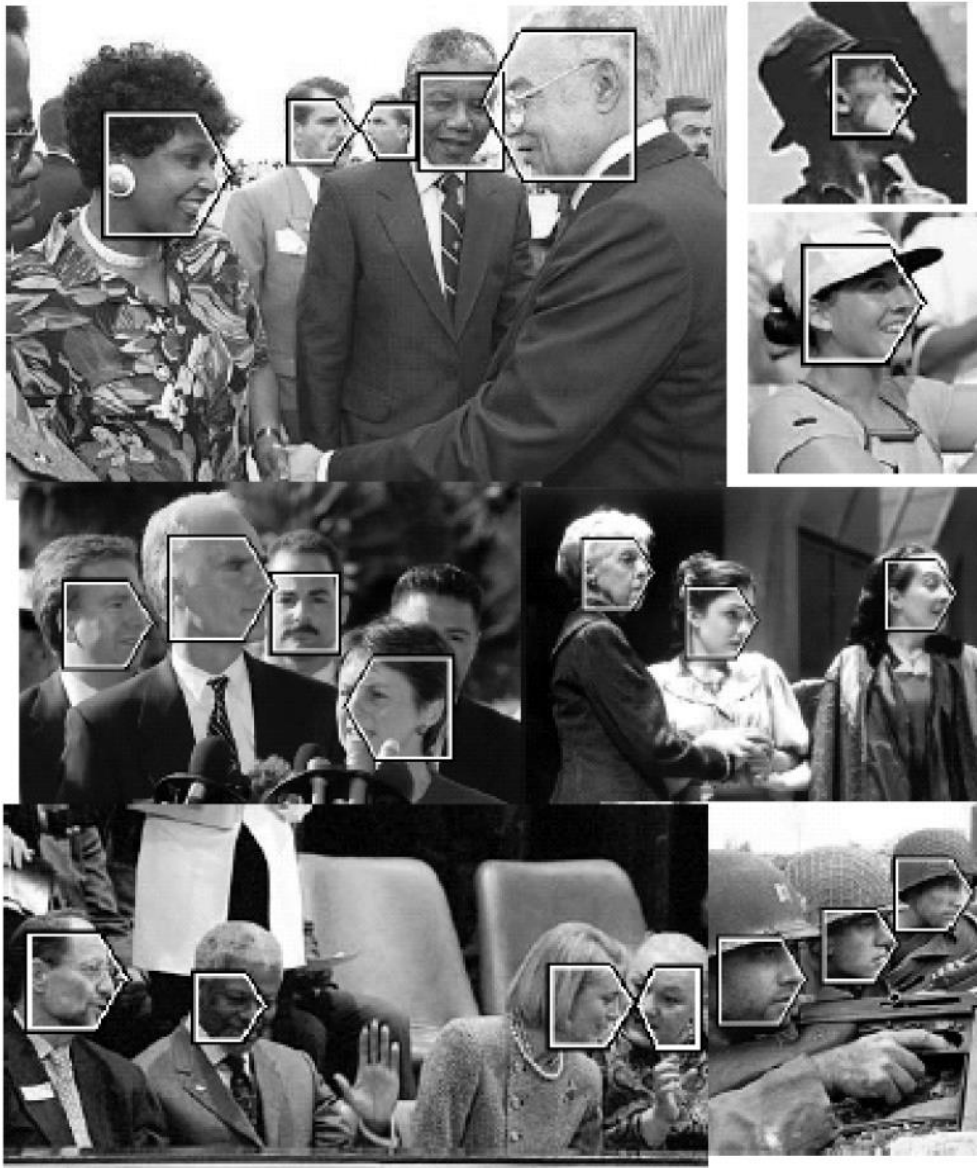


What are the desired outcomes?

What are the inputs (data)?

What are the learning paradigms?

Face Detection



What are the desired outcomes?

What are the inputs (data)?

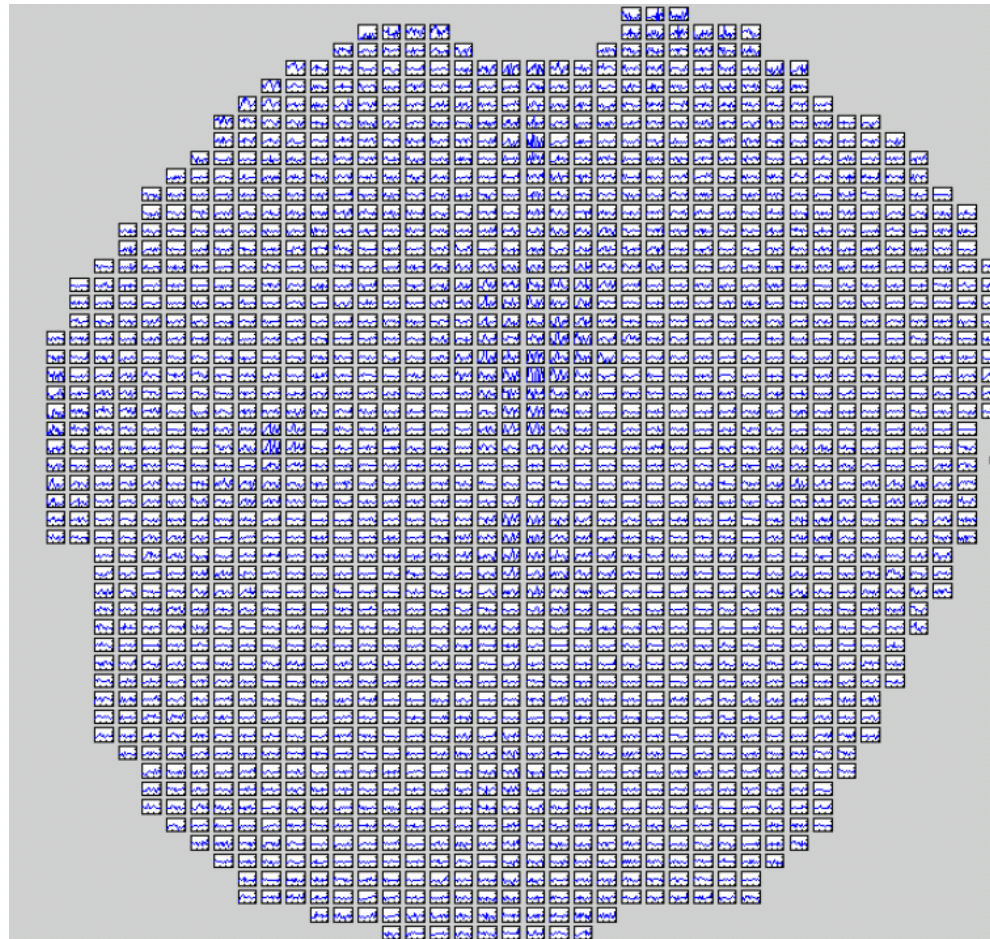
What are the learning paradigms?

Understanding brain activity

What are the desired outcomes?

What are the inputs (data)?

What are the learning paradigms?

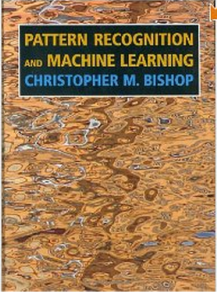


Reading
a noun
(vs verb)

[Rustandi et al.,
2005]

Product Recommendation

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CHRISTOPHER M. BISHOP

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Christopher M. Bishop (Author)
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What are the desired outcomes?

What are the inputs (data)?

What are the learning paradigms?

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 +  + 

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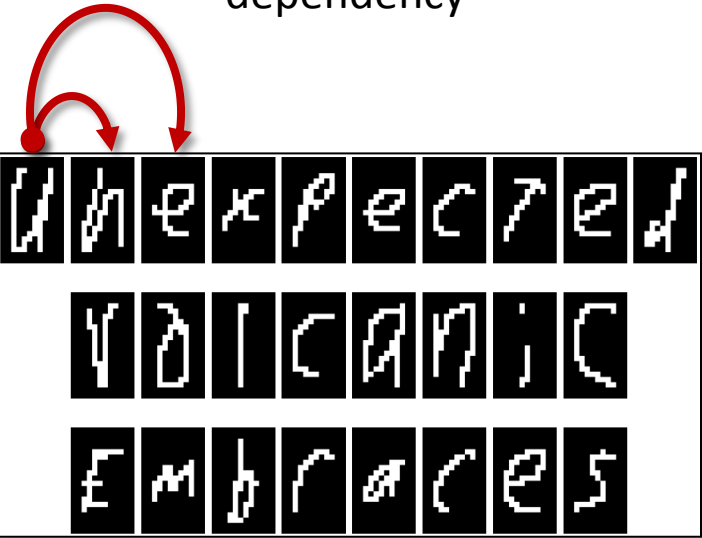
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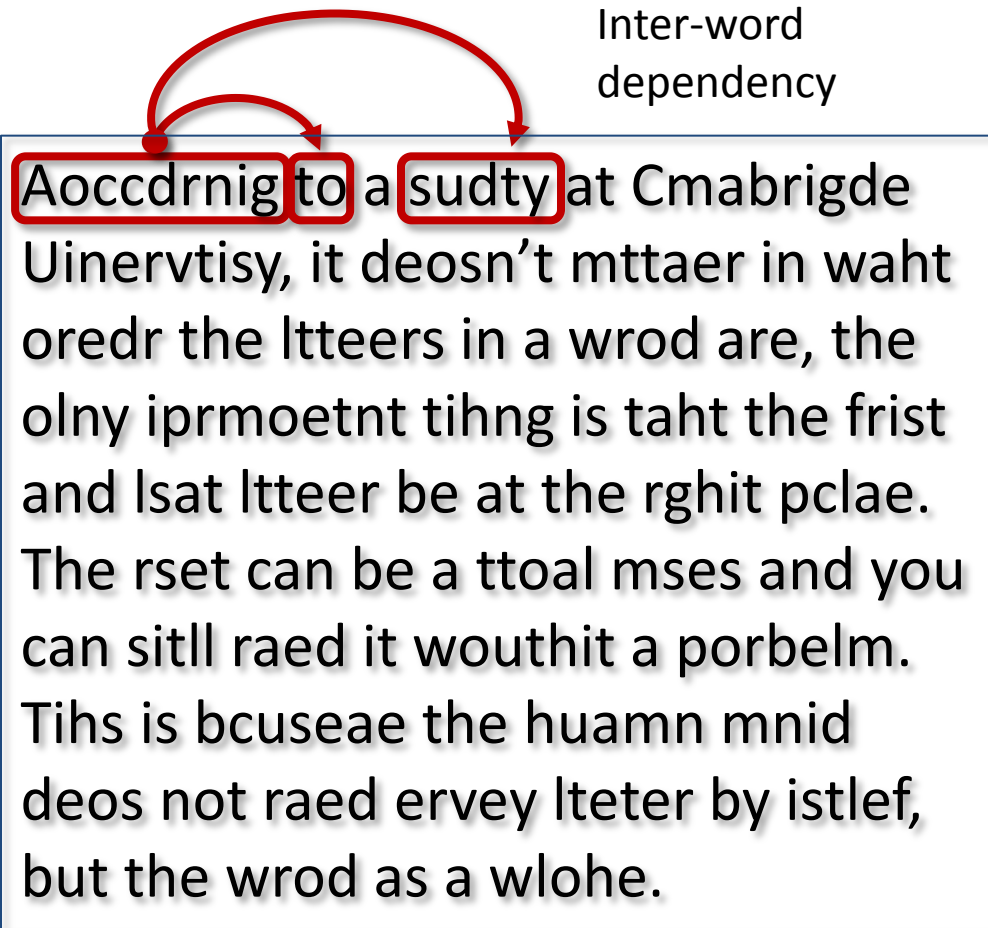
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Handwritten digit recognition/text annotation

Inter-character
dependency



Inter-word
dependency



What are the desired outcomes?

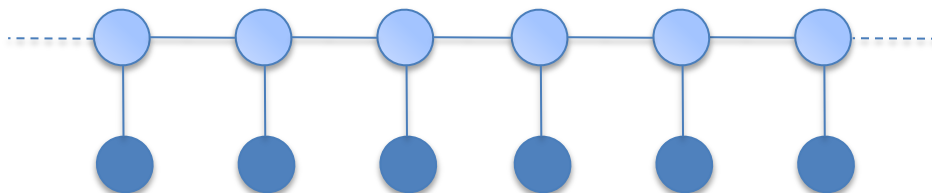
What are the inputs (data)?

What are the learning paradigms?

Similar problem: speech recognition

Models

Hidden Markov Models



Text

“Machine Learning is the preferred method for speech recognition ...”



Audio signals

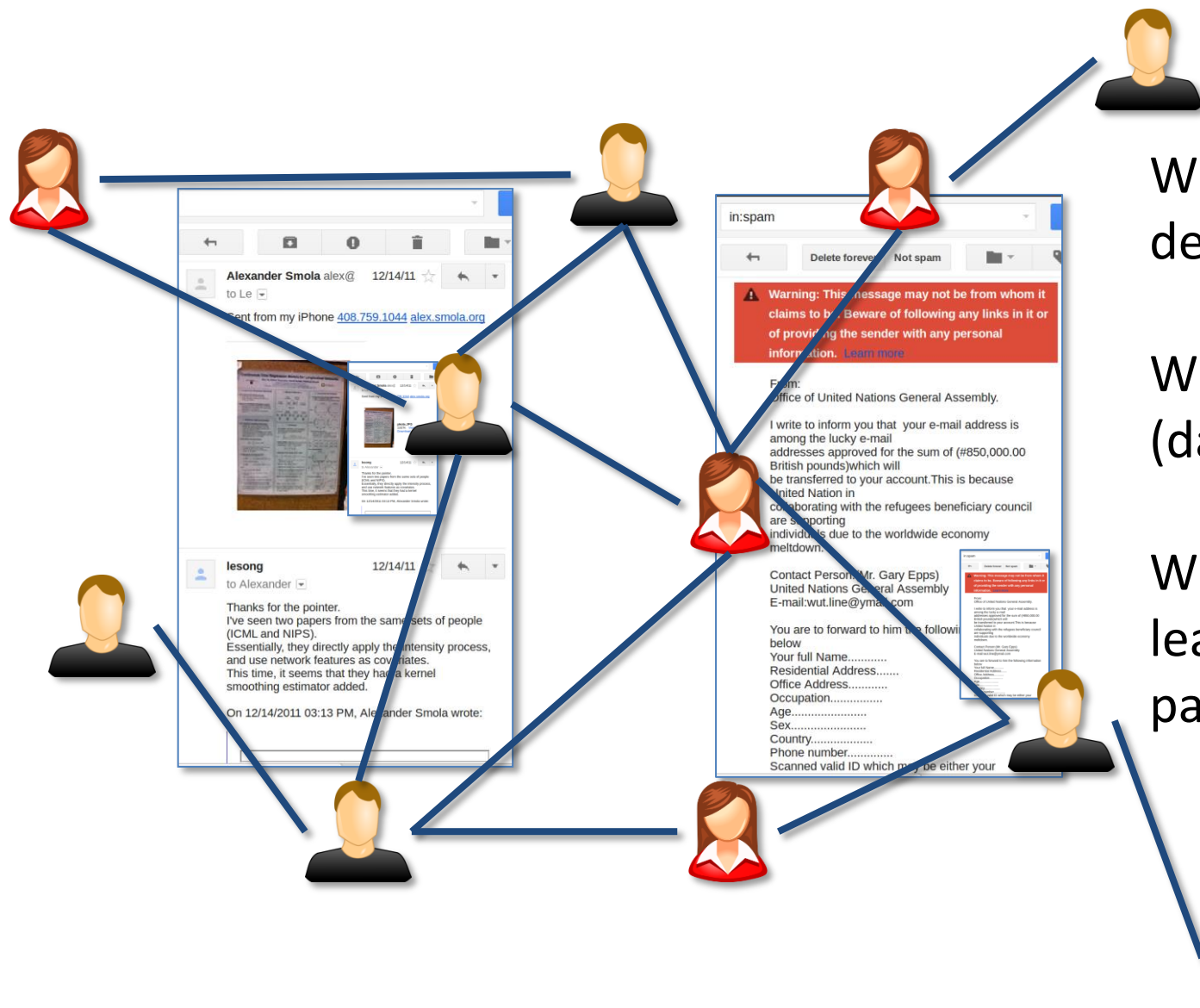


Similar problem: bioinformatics



Where is the gene?

Spam Filtering

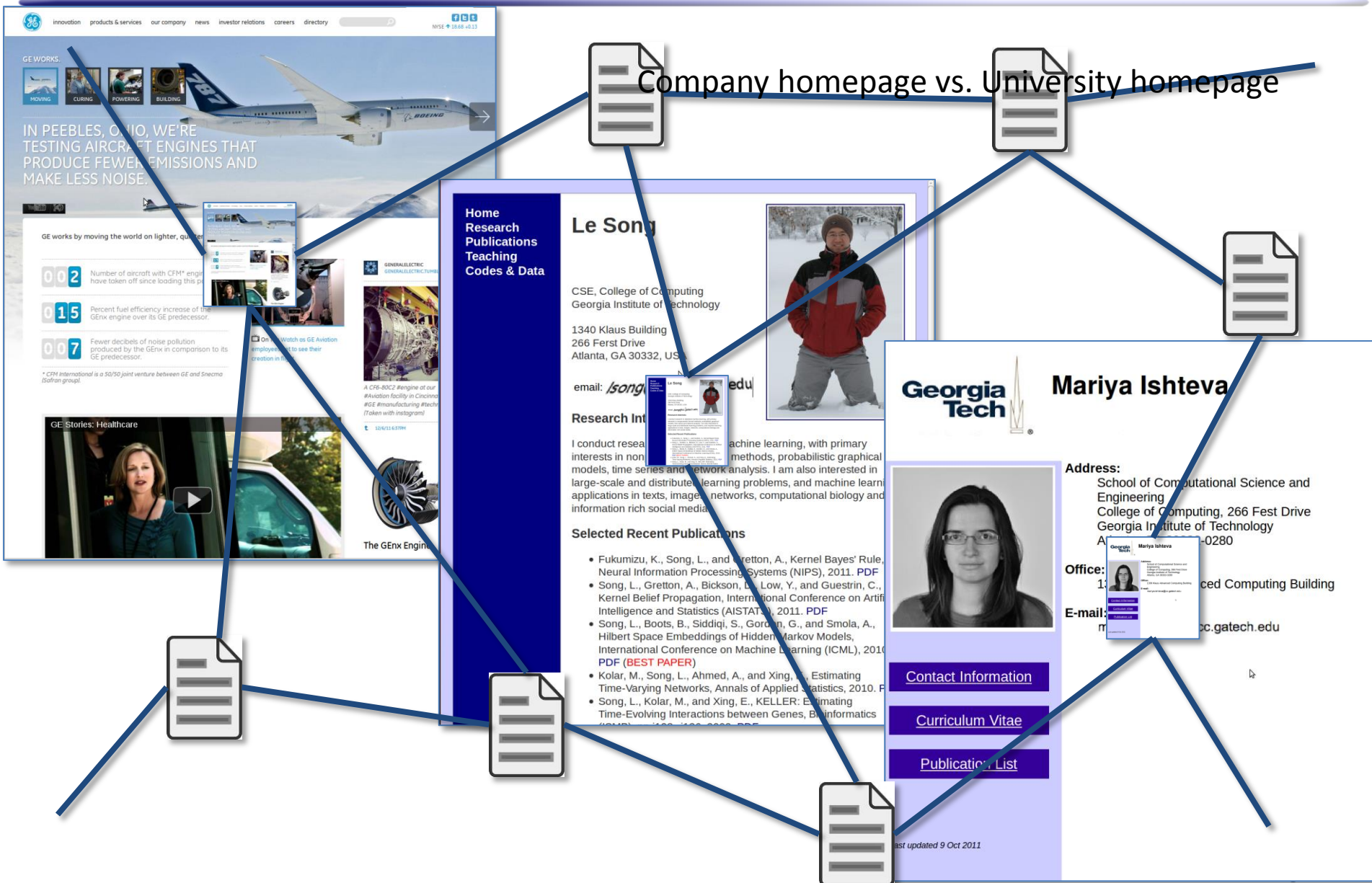


What are the desired outcomes?

What are the inputs (data)?

What are the learning paradigms?

Similar problem: webpage classification



Robot Control

- Now cars can find their own ways!



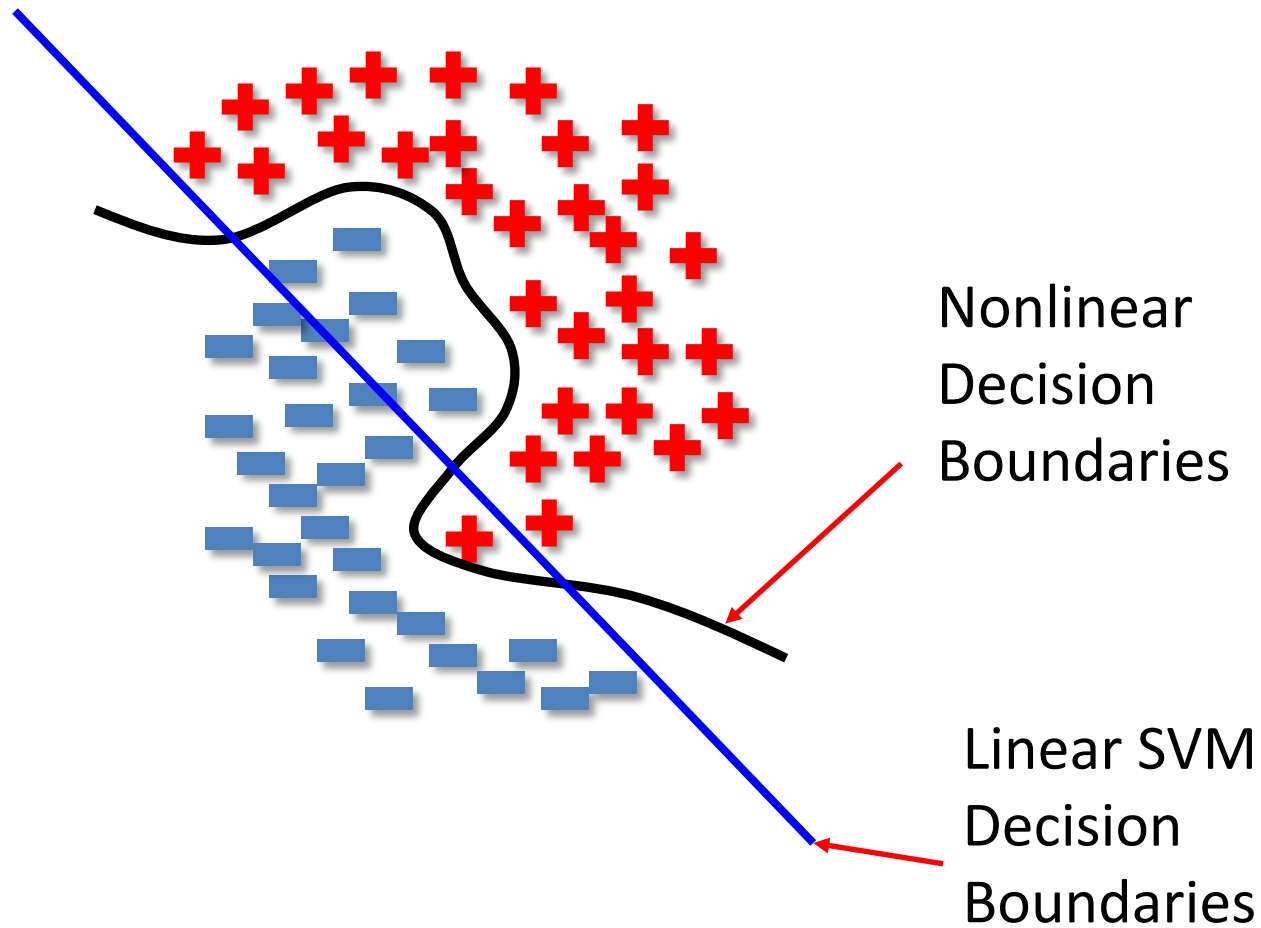
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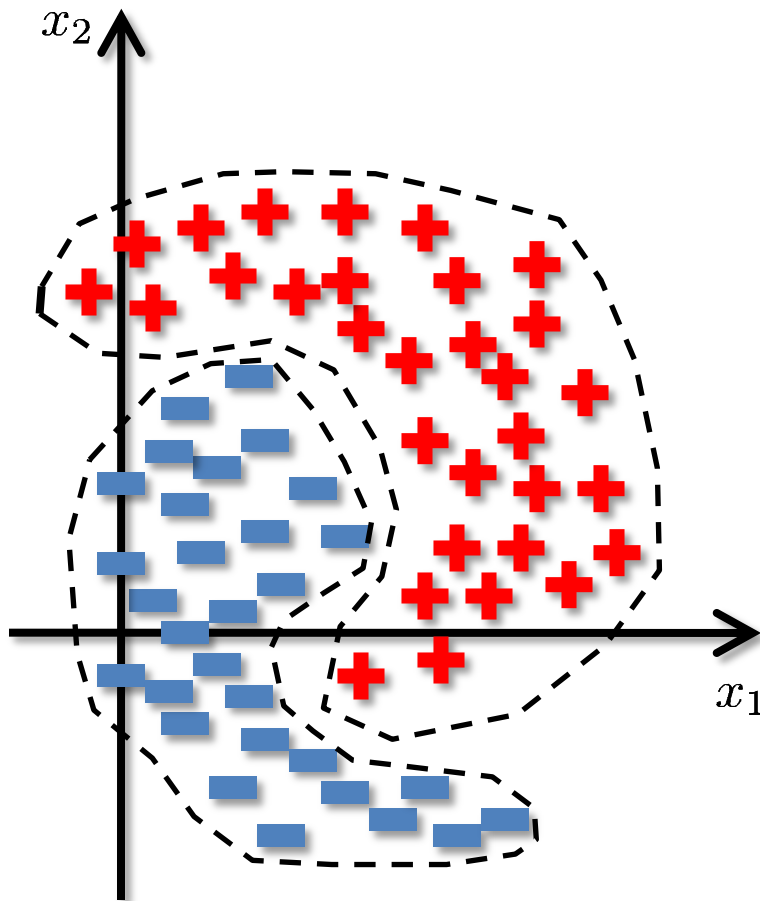
What are the learning paradigms?



Nonlinear classifier



Nonconventional clusters



Need more advanced methods, such as kernel methods or spectral clustering to work

Syllabus

- Cover a number of most commonly used machine learning algorithms in sufficient amount of details.
- Explain how to code these algorithms in Matlab and make them work on real datasets
- Organization
 - Unsupervised learning (data exploration)
 - Clustering, dimensionality reduction, density estimation
 - Supervised learning (predictive models)
 - Classifications, regressions
 - Nonlinear models, combine models, model selection

Prerequisites

- Probabilities
 - Distributions, densities, marginalization, conditioning
- Basic statistics
 - mean, variance, covariance, maximum likelihood estimation
- Linear algebra
 - Vector, matrix, inversion, eigen-decomposition
- Algorithms
 - Basic data structures, computational complexity
- Programming
 - Knowledge of one form of programming language

Textbooks

- Textbooks:
 - Machine Learning: a Probabilistic Perspective, by Kevin Patrick Murphy
 - Pattern Recognition and Machine Learning, Chris Bishop
- Programing guide:
 - <http://www.mathworks.com/help/matlab/>

Grading

- 6 assignments (60%): approximately 1 assignment every 4 lectures (start early)
- Midterm exam (20%): Testing basic matlab program questions and basic understanding of machine learning algorithms
- Final project (20%): program with matlab, submit your results to a classification competition:
 - <http://www.kaggle.com/c/cifar-10>
 - Your accuracy score there will be your project score
- Advanced option: Imagenet classification:
 - <http://www.image-net.org/challenges/LSVRC/2013/>
 - Your accuracy score x 4 will be your project score

Homeworks

- Zero credit after each deadline
- All homeworks **must be handed in**, even for zero credit
- Collaboration
 - You may **discuss** the questions
 - Each student writes their own answers
 - Write on your homework anyone with whom you collaborate
 - Each student must write their own codes for the programming part

Staff

- Instructor: Le Song, Klaus 1340
- TA: Wei Da, Klaus 1305, Thiaga Rajan, Klaus 1305
- Guest Lecturer: TBD
- Administrative Assistant: Mimi Haley, Klaus 1321
- Mailing list: TBD
- More information:

<http://www.cc.gatech.edu/~lsong/teaching/CX4240Spring14.html>

Today

- Probabilities
- Independence
- Conditional Independence

Random Variables (RV)

- Data may contain many different attributes
 - Age, grade, color, location, coordinate, time ...
- Upper-case for random variables (eg. X, Y), lower-case for values (eg. x, y)
- $P(X)$ for distribution, $p(X)$ for density
 - $Val(X)$ = possible values of random variable X
 - For discrete (categorical): $\sum_{i=1 \dots |Val(X)|} P(X = x_i) = 1$
 - For continuous: $\int_{Val(X)} p(X = x) dx = 1$
 - $P(x) \geq 0$
- Shorthand: $P(x)$ for $P(X = x)$

Interpretations of probability

- Frequentists
 - $P(x)$ is the frequency of x in the limit
 - Many arguments against this interpretation
 - What is the frequency of the event “it will rain tomorrow?”
- Subjective interpretation
 - $P(x)$ is my degree of belief that x will happen
 - What does “degree of belief mean”?
 - If $P(x) = 0.8$, then I am willing to bet
- For this class, we don't care the type of interpretation.

Conditional probability

- After we have seen x , how do we feel y will happen?
- $P(y|x)$ means $P(Y = y|X = x)$
- A conditional distribution are a family of distributions
 - For each $X = x$, it is a distribution $P(Y|x)$

Two of the most important rules: I. The chain rule

- $P(y, x) = P(y|x)P(x)$
- More generally:
 - $P(x_1, x_2, \dots, x_k) = P(x_1)P(x_2|x_1) \dots P(x_k|x_{k-1}, \dots, x_2, x_1)$

Two of the most important rules: II. Bayes rule

likelihood Prior

posterior Normalization constant

$$\bullet \quad P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{\sum_{x \in \text{Val}(X)} P(x,y)}$$

- More generally, additional variable z :

$$\bullet \quad P(y|x, z) = \frac{P(x|y,z)P(y|z)}{P(x|z)}$$

Independence

- X and Y independent, if $P(Y|X) = P(Y)$
 - $P(Y|X) = P(Y) \rightarrow (X \perp Y)$
- Proposition: X and Y independent if and only if $P(X, Y) = P(X)P(Y)$

Conditional independence

- Independence is rarely true; conditional independence is more prevalent
- X and Y conditionally independent given Z if
$$P(Y|X, Z) = P(Y|Z)$$
 - $P(Y|X, Z) = P(Y|Z) \rightarrow (X \perp Y | Z)$
- $(X \perp Y | Z)$ if and only if $P(X, Y | Z) = P(X | Z)P(Y | Z)$

Joint distribution, marginalization

- Two random variables – Grades (G) & Intelligence (I)

• $P(G, I) =$

G \ I	VH	H
A	0.7	0.1
B	0.15	0.05

- For n binary variables, the table (multiway array) gets really big
 - $P(X_1, X_2, \dots, X_n)$ has 2^n entries!
- Marginalization – Compute marginal over a single variable
 - $P(G = B) = P(G = B, I = VH) + P(G = B, I = H) = 0.2$

Marginalization – the general case

- Compute marginal distribution $P(X_i)$ from $P(X_1, X_2, \dots, X_i, X_{i+1}, \dots, X_n)$

$$P(X_1, X_2, \dots, X_i) = \sum_{x_{i+1}, \dots, x_n} P(X_1, X_2, \dots, X_i, x_{i+1}, \dots, x_n)$$

$$P(X_i) = \sum_{x_1, \dots, x_{i-1}} P(x_1, \dots, x_{i-1}, X_i)$$

If binary variables, need to sum over 2^{n-1} terms!

Example problem

- Estimate the probability θ of landing in heads using a biased coin
- Given a sequence of N independently and identically distributed (iid) flips
 - Eg., $D = \{x_1, x_2, \dots, x_N\} = \{1, 0, 1, \dots, 0\}, x_i \in \{0, 1\}$
- Model: $P(x|\theta) = \theta^x(1 - \theta)^{1-x}$
 - $P(x|\theta) = \begin{cases} 1 - \theta, & \text{for } x = 0 \\ \theta, & \text{for } x = 1 \end{cases}$
- Likelihood of a single observation x_i ?
 - $P(x_i|\theta) = \theta^{x_i}(1 - \theta)^{1-x_i}$



Frequentist Parameter Estimation

- Frequentists think of a parameter as a **fixed, unknown constant**, not a random variable
- Hence different “objective” estimators, instead of Bayes’ rule
 - These estimators have different properties, such as being “unbiased”, “minimum variance”, etc.
- A very popular estimator is the **maximum likelihood estimator (MLE)**, which is simple and has good statistical properties
 - $\theta = \operatorname{argmax}_{\theta} P(D|\theta) = \operatorname{argmax}_{\theta} \prod_{i=1}^N P(x_i|\theta)$

MLE for Biased Coin

- Objective function, log likelihood
 - $l(\theta; D) = \log P(D|\theta) = \log \theta^{n_h} (1 - \theta)^{n_t} = n_h \log \theta + (N - n_h) \log(1 - \theta)$
- We need to maximize this w.r.t. θ
- Take derivatives w.r.t. θ
- $\frac{\partial l}{\partial \theta} = \frac{n_h}{\theta} - \frac{(N - n_h)}{1 - \theta} = 0 \Rightarrow \hat{\theta}_{MLE} = \frac{n_h}{N} \text{ or } \hat{\theta}_{MLE} = \frac{1}{N} \sum_i x_i$

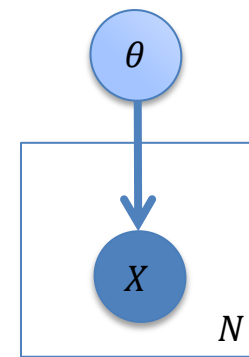
Bayesian Parameter Estimation

- Bayesian treat the unknown parameters as a random variable, whose **distribution** can be inferred using Bayes rule:

- $$P(\theta | D) = \frac{P(D|\theta)P(\theta)}{P(D)} = \frac{P(D|\theta)P(\theta)}{\int P(D|\theta)P(\theta)d\theta}$$

- The crucial equation can be written in words

- $$\text{Posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{marginal likelihood}}$$



- For iid data, the likelihood is $P(D|\theta) = \prod_{i=1}^N P(x_i|\theta)$

- $$\prod_{i=1}^N \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum_i x_i} (1 - \theta)^{\sum_i 1-x_i} = \theta^{\#head} (1 - \theta)^{\#tail}$$

- The prior $P(\theta)$ encodes our prior knowledge on the domain

- Different prior $P(\theta)$ will end up with different estimate $P(\theta|D)$!

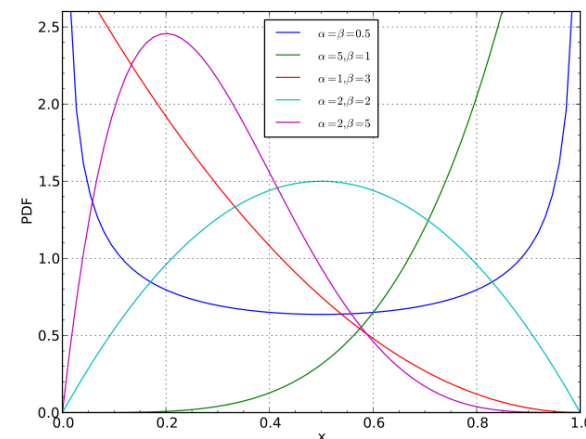
Bayesian estimation for biased coin

- Prior over θ , Beta distribution

- $P(\theta; \alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$
- When x is discrete $\Gamma(x+1) = x\Gamma(x) = x!$

- Posterior distribution of θ

- $P(\theta|x_1, \dots, x_N) = \frac{P(x_1, \dots, x_N|\theta)P(\theta)}{P(x_1, \dots, x_N)} \propto$
 $\theta^{n_h} (1-\theta)^{n_t} \theta^{\alpha-1} (1-\theta)^{\beta-1} = \theta^{n_h+\alpha-1} (1-\theta)^{n_t+\beta-1}$
- α and β are hyperparameters and correspond to the number of “virtual” heads and tails (pseudo counts)



Bayesian Estimation for Bernoulli

- Posterior distribution θ

- $$P(\theta|x_1, \dots, x_N) = \frac{P(x_1, \dots, x_N|\theta)P(\theta)}{P(x_1, \dots, x_N)} \propto$$
$$\theta^{n_h}(1-\theta)^{n_t}\theta^{\alpha-1}(1-\theta)^{\beta-1} = \theta^{n_h+\alpha-1}(1-\theta)^{n_t+\beta-1}$$

- Posterior mean estimation:

- $$\theta_{bayes} = \int \theta P(\theta|D)d\theta = C \int \theta \times \theta^{n_h+\alpha-1}(1-\theta)^{n_t+\beta-1}d\theta =$$
$$\frac{(n_h+\alpha)}{N+\alpha+\beta}$$

- Prior strength: $A = \alpha + \beta$

- A can be interpreted as an imaginary dataset